LAB 8: State Transition Diagram

Smart Waste Collection Management System

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1. Objective

To model the life cycle of the important objects of the Smart Waste Collection Management System as UML state transition (state machine) diagrams. Each diagram shows the states of an object, the events and guard conditions that move it from one state to another, and the actions performed. The behaviour is taken from SRS v1.0.

2. State transition diagrams included

Figure	Object	States	SRS reference
Diagram 1 (page 2)	Bin	Normal, Near Full, Scheduled for Collection, Overflowing, Collected	4.1, 4.2, 5.5 (REQ-AL-1, REQ-AL-2, REQ-RO-1)
Diagram 2 (page 3)	Sensor Node	Sleeping, Measuring, Transmitting, Error Handling	4.1 (REQ-BM-1 to 3)
Diagram 3 (page 4)	Route	Computing, Generated, Adjusting, Published, In Progress, Completed	4.3 (REQ-RO-1 to 3)
Diagram 4 (page 5)	Citizen Complaint	Pending, Under Review, Pickup Scheduled, Resolved	4.5, 5.5 (REQ-CF-1 to 3)

3. Main transitions

- **Bin:** becomes Near Full when a reading is at or above the threshold, then Scheduled once a route includes it, and returns to Normal after it is collected and the fill-level is reset.
- **Sensor Node:** wakes every 15 minutes to measure; a valid reading is transmitted, a read error is retried and logged until the retry limit is reached.
- **Route:** is computed on request, may be adjusted by the administrator, is pushed to the driver, and completes when all bins are collected.
- **Complaint:** stays Pending for the citizen until the linked bin is collected, when it is automatically set to Resolved.

4. Notation used

Symbol	Meaning
Filled circle	Initial state (start of the life cycle)
Rounded rectangle	State, with its entry / do actions below the name
Arrow with label	Transition: event [guard condition] / action
Small diamond	Choice (decision) pseudo-state
Loop on a state	Self transition (the object stays in the same state)
Circle with filled centre	Final state (end of the life cycle)

State Transition Diagram 1 - Bin

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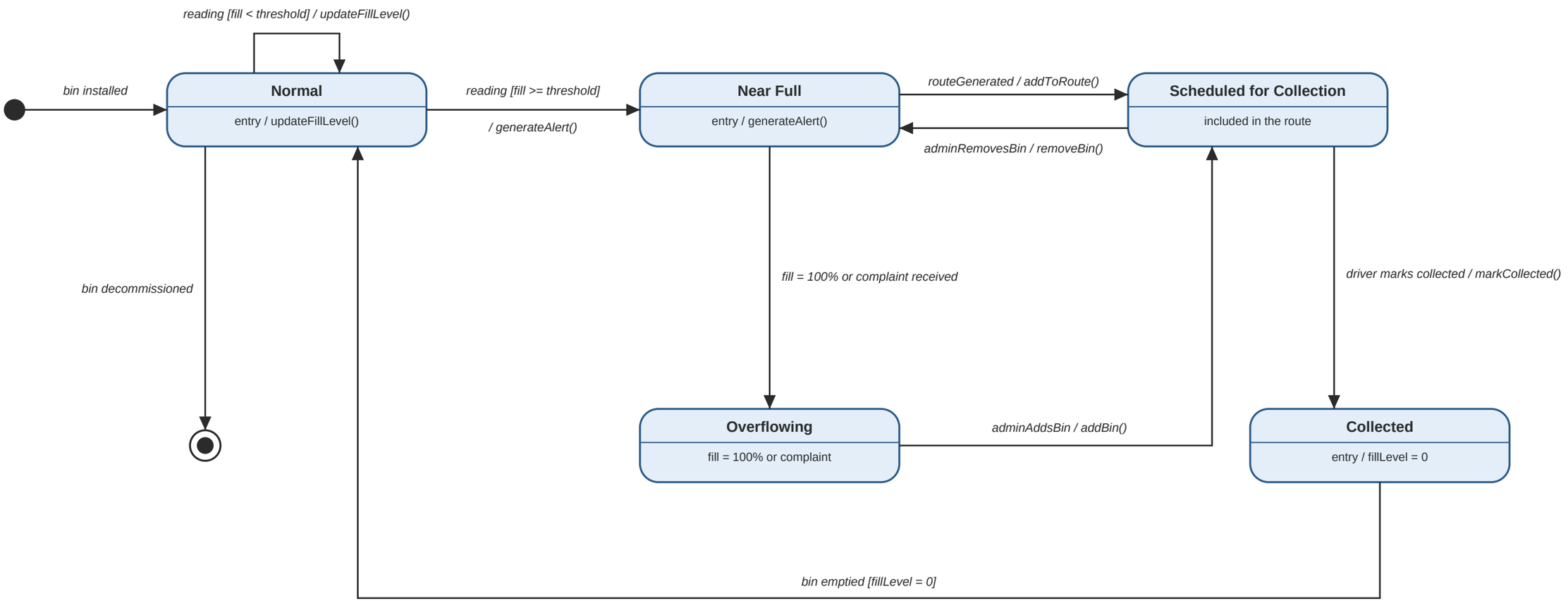

Bin states follow the fill-level against the alert threshold (REQ-AL-1, REQ-AL-2) and the collection rule in SRS 5.5.

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State Transition Diagram 2 - Sensor Node

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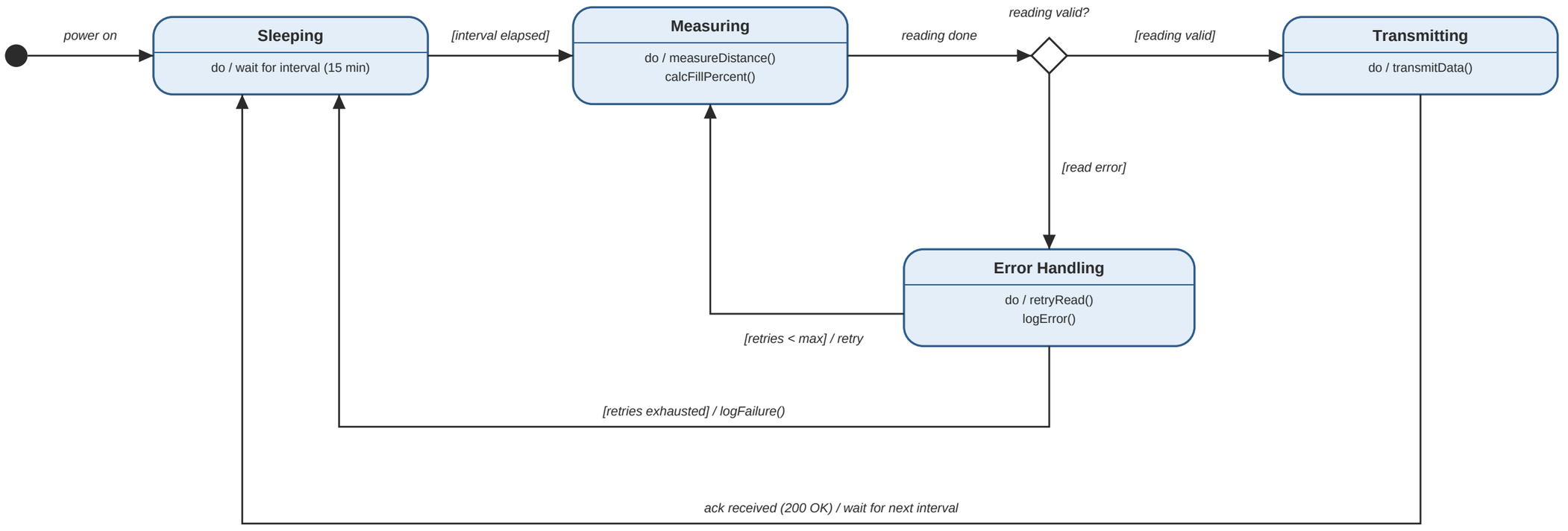

Fixed 15-minute interval and error retry/logging follow REQ-BM-1 to REQ-BM-3.

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State Transition Diagram 3 - Route

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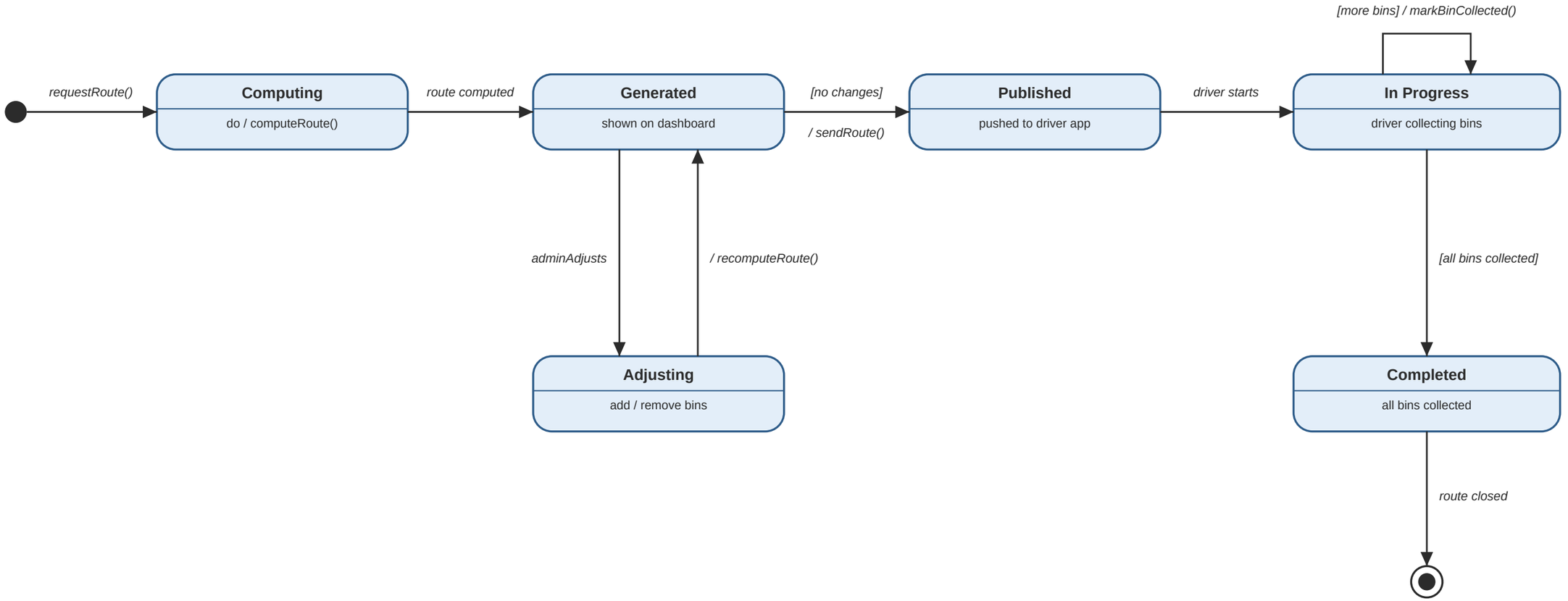

Route computation must finish within 10 seconds for up to 200 bins (REQ-RO-2); admin edits follow REQ-RO-3.

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State Transition Diagram 4 - Citizen Complaint

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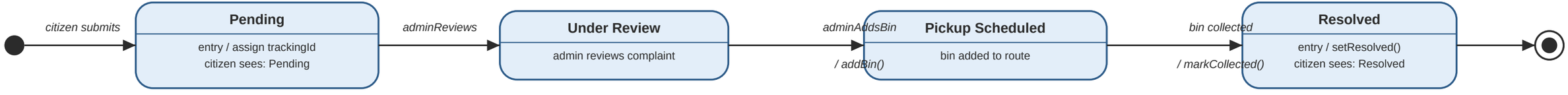

The citizen sees the status Pending while the complaint is Pending, Under Review or Pickup Scheduled (REQ-CF-3).
It is set to Resolved automatically when the corresponding bin is recorded as collected (SRS 5.5).

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